

PORT EDWARD, South Africa

WMO: 685870

Lat: 31.0622S

Lon: 30.2214E

Elev: 11

StdP: 101.19

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	12.6	0.0	3.8	16.3	2.1	4.4	15.9	10.8	18.8	9.5	18.5	2.4	320	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.7	27.9	24.2	27.2	23.9	26.7	23.4	25.5	26.6	24.6	26.2	24.0	25.8	4.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.3	26.2	24.2	19.1	25.6	23.3	18.1	25.3	78.4	26.6	74.8	26.2	72.1	25.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.4	7.4	8.9	29.9	1.4	1.5	7.9	30.9	7.1	31.8	6.3	32.6	5.3	33.6		
(5)			5.5	25.4	0.8	0.9	4.9	26.1	4.5	26.6	4.0	27.2	3.4	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.5	23.4	23.8	23.1	21.3	19.8	18.1	17.5	18.1	18.9	19.5	20.9	22.4
	DBStd	2.70	1.53	1.48	1.49	1.79	1.56	1.50	1.50	1.70	1.85	1.79	1.80	1.69
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	114	0	0	0	2	5	22	34	24	15	9	2	0
	CDD10.0	3848	417	387	406	338	303	242	231	252	266	296	328	383
	CDD18.3	921	158	153	148	90	49	13	7	18	31	47	80	125
	CDH23.3	2929	714	791	584	193	71	17	12	12	15	23	123	375
CDH26.7	128	37	51	17	3	3	1	2	1	1	0	3	9	
Wind	WSAvg	3.4	3.4	3.2	3.4	3.3	3.1	3.0	3.1	3.6	3.8	4.0	3.8	3.6
Precipitation	PrecAvg	1101	126	104	156	82	46	31	48	45	88	114	139	138
	PrecMax	1698	307	257	388	159	100	147	169	148	233	239	251	222
	PrecMin	813	53	37	47	25	4	2	0	8	18	27	61	49
	PrecStd	256	62	54	85	42	27	35	53	39	53	56	59	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	28.9	28.0	27.1	26.6	25.0	24.7	24.7	24.7	25.0	26.4	27.7
		MCWB	25.6	25.8	23.7	23.0	19.0	16.6	15.8	18.4	19.8	20.6	22.3	24.7
	2%	DB	27.6	27.9	27.2	26.0	25.0	23.6	23.0	23.1	23.4	24.0	25.4	26.6
		MCWB	24.6	24.5	23.6	22.5	20.2	17.2	17.0	18.5	19.9	20.5	22.3	23.9
	5%	DB	27.0	27.3	26.6	25.2	24.0	22.7	22.1	22.3	22.8	23.2	24.7	26.0
		MCWB	24.0	24.1	23.3	21.7	19.6	17.0	17.2	18.2	19.4	19.9	21.9	23.2
	10%	DB	26.3	26.7	26.0	24.5	23.2	21.8	21.2	21.6	22.1	22.6	24.0	25.3
		MCWB	23.3	23.5	22.8	21.1	19.3	16.6	16.7	17.9	18.9	19.4	21.2	22.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.4	25.7	23.8	22.1	20.6	19.9	21.1	21.6	21.8	24.2	26.0
		MCDB	27.6	27.2	26.6	25.9	24.5	22.4	21.7	22.6	23.5	23.4	25.1	26.6
	2%	WB	25.4	25.2	24.7	23.0	21.1	19.1	18.8	19.9	20.9	21.1	23.2	24.7
		MCDB	26.9	26.6	26.0	25.2	23.8	22.2	21.6	22.1	22.6	22.9	24.5	25.6
	5%	WB	24.4	24.3	23.8	22.3	20.5	18.2	18.0	19.1	20.2	20.4	22.4	23.7
		MCDB	26.3	26.3	25.8	24.3	23.2	21.5	21.0	21.7	22.3	22.4	23.9	24.9
	10%	WB	23.6	23.6	23.1	21.5	19.8	17.4	17.2	18.4	19.5	19.8	21.5	23.0
		MCDB	25.7	25.8	25.3	23.7	22.6	20.9	20.4	21.2	21.8	21.9	23.3	24.6
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.7	4.7	5.4	6.5	7.4	7.3	6.1	5.3	4.7	4.5	4.4
		MCDBR	5.2	5.3	5.5	6.1	7.6	8.8	8.8	7.2	6.0	5.3	5.3	5.1
		MCWBR	3.5	3.5	3.8	4.3	5.3	7.0	7.0	5.8	4.5	4.3	4.1	3.7
	5% WB	MCDBR	5.2	5.0	5.0	5.5	6.3	7.9	8.0	6.7	5.4	5.1	4.9	4.6
		MCWBR	3.7	3.7	3.6	4.1	5.1	6.9	6.7	5.7	4.5	4.1	3.8	3.5
Clear-Sky Solar Irradiance	taub	0.446	0.442	0.430	0.396	0.372	0.354	0.364	0.428	0.470	0.473	0.448	0.447	
	taud	2.327	2.344	2.358	2.417	2.447	2.471	2.430	2.238	2.134	2.171	2.282	2.310	
	Ebn at Noon	898	882	857	833	804	797	803	786	804	843	890	900	
	Edh at Noon	136	130	121	103	89	82	89	121	148	153	142	139	
All-Sky Solar Radiation	RadAvg	5.98	5.74	4.92	4.02	3.34	2.99	3.20	3.78	4.43	4.94	5.53	5.80	
	RadStd	0.51	0.44	0.27	0.27	0.17	0.15	0.21	0.24	0.30	0.46	0.40	0.42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page